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Supplementary Materials for

Multicolor three-photon fluorescence imaging with single-wavelength excitation deep in mouse brain

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Other Supplementary Material for this manuscript includes the following:

(available at advances.sciencemag.org/cgi/content/full/7/12/eabf3531/DC1)

Movies S1 to S3

Supplemental Note 1: ***Signal-to-background ratio (SBR) calculation***

We calculate the signal-to-background ratio (SBR) when a mixture of 2PE and 3PE fluorescence is obtained. The SBRs for 2PE (SBR_{2P}), 3PE (SBR_{3P}), and a mix of 2PE and 3PE (SBR_{total}) are defined as

$$SBR_{2P} = \frac{S_{2P}}{B_{2P}} \tag{3}$$

$$SBR_{3P} = \frac{S_{3P}}{B_{3P}} \tag{4}$$

$$SBR_{total} = \frac{S_{2P}+S_{3P}}{B_{2P}+B_{3P}} \tag{5}$$

where S_{2P}, S_{3P}, B_{2P} and B_{3P} are 2PE signal, 3PE signal, 2PE background and 3PE background, respectively. Because out-of-focus background fluorescence in 3PE is negligible for most practical imaging depths (*e.g.*, within 1 to 2 mm in the mouse brain) (*25*), SBR_{total} can be approximated as

$$SBR_{total} \approx \frac{S_{2P}+S_{3P}}{B_{2P}} = \frac{S_{2P}}{B_{2P}}\left(1+\frac{S_{3P}}{S_{2P}}\right) = SBR_{2P}\left(1+\frac{S_{3P}}{S_{2P}}\right) \tag{6}$$

Hence, the ratio of SBR_{total} to SBR_{2P} is approximately

$$\frac{SBR_{total}}{SBR_{2P}} \approx 1+\frac{S_{3P}}{S_{2P}} \tag{7}$$

Based on the Eqs. (1) and (2) in the Materials and Method section, S_{3P}/S_{2P} is calculated as

$$\frac{S_{3P}}{S_{2P}} = \frac{3.5\pi(NA)^2}{12}\frac{g_p^{(3)}}{g_p^{(2)}}\frac{\sigma_3}{\sigma_2}\frac{\langle P(t)\rangle}{(f\tau)\lambda^2} \tag{8}$$

where $g_p^{(n)}$ is the nth-order temporal coherence of the excitation source, σ_n is the n-photon action cross section, NA is the numerical aperture of the objective lens, $\langle P(t)\rangle$ is the time-averaged excitation photon flux (photons/s), λ is the excitation wavelength in vacuum, f is the laser repetition rate, and τ is the laser pulse duration (full width at half maximum). With the known 2P and 3P cross sections, SBR_{total}/SBR_{2P} can be calculated for various excitation conditions. Based on the measured 2P and 3P action cross sections (**Fig. 1**), we calculated SBR_{total}/SBR_{2P} at various excitation wavelengths for all the fluorophores in this study (**Figs. S6** and **S7**). A typical 3PM condition is used for the calculation: NA of 0.8 and a pulse duration of 60 fs. For **Fig. S6**, a pulse energy of 1.0 nJ at the focus was used, except for 1260 nm where 0.4 nJ was used as in the *in vivo* experiment in **Fig. 2**. For **Fig. S7**, pulse energies used for calculation are shown in the table in **Fig. S7**.

Supplemental Note 2: ***Analysis of the photobleaching quantum yield***

In a simplified model, the photobleaching process can be described by a single constant, the photobleaching quantum yield (Φ_b).

$$\frac{dn}{dt} = - nE \times \Phi_b \tag{9}$$

where n is the number of fluorescent molecules at the focus of the excitation beam and E is the excitation rate. From Eq. (9), $n(t)$ is calculated as

$$n(t) = n_0 \exp(-E\Phi_b t) \tag{10}$$

where n_0 is the number of fluorescent molecules at $t = 0$. The number of emitted fluorescence photons as a function of time, $F(t)$, is proportional to $n(t)$,

$$F(t) = F_0 \exp(-E\Phi_b t) \tag{11}$$

where F_0 is the number of emitted photons at $t = 0$. The fluorescence decay time constant (τ) in Eq. (11) is then $\tau = 1/E\Phi_b$. Therefore, the photobleaching quantum yield Φ_b can be obtained as

$$\Phi_b = \tau / E \tag{12}$$

The excitation rate E can be calculated based on the 3PE cross sections (Fig. 1), excitation power and pulse durations. For the comparison of Φ_b at 1340 nm and 1650 nm, *i.e.*, Φ_b(1340 nm)/Φ_b(1650 nm), only the relative excitation rate, *i.e.*, E(1340 nm)/E(1650 nm), is needed. In our experiments shown in Figs. 5 and S11, the excitation rates at 1340 nm with excitation pulse energy of 1.3 nJ and 1.1 nJ are, respectively, ~4.8 and ~2.9 times larger than that at 1650 nm (2.0 nJ), when the excitation pulse durations and the effective attenuation lengths (see ref. 14 in the main text) are taken into account. From the relative excitation rates and the time constants of the photobleaching traces (Figs. 5B,C and S11B), the relative photobleaching quantum yields were calculated and plotted in Figs. 5D and S11C.

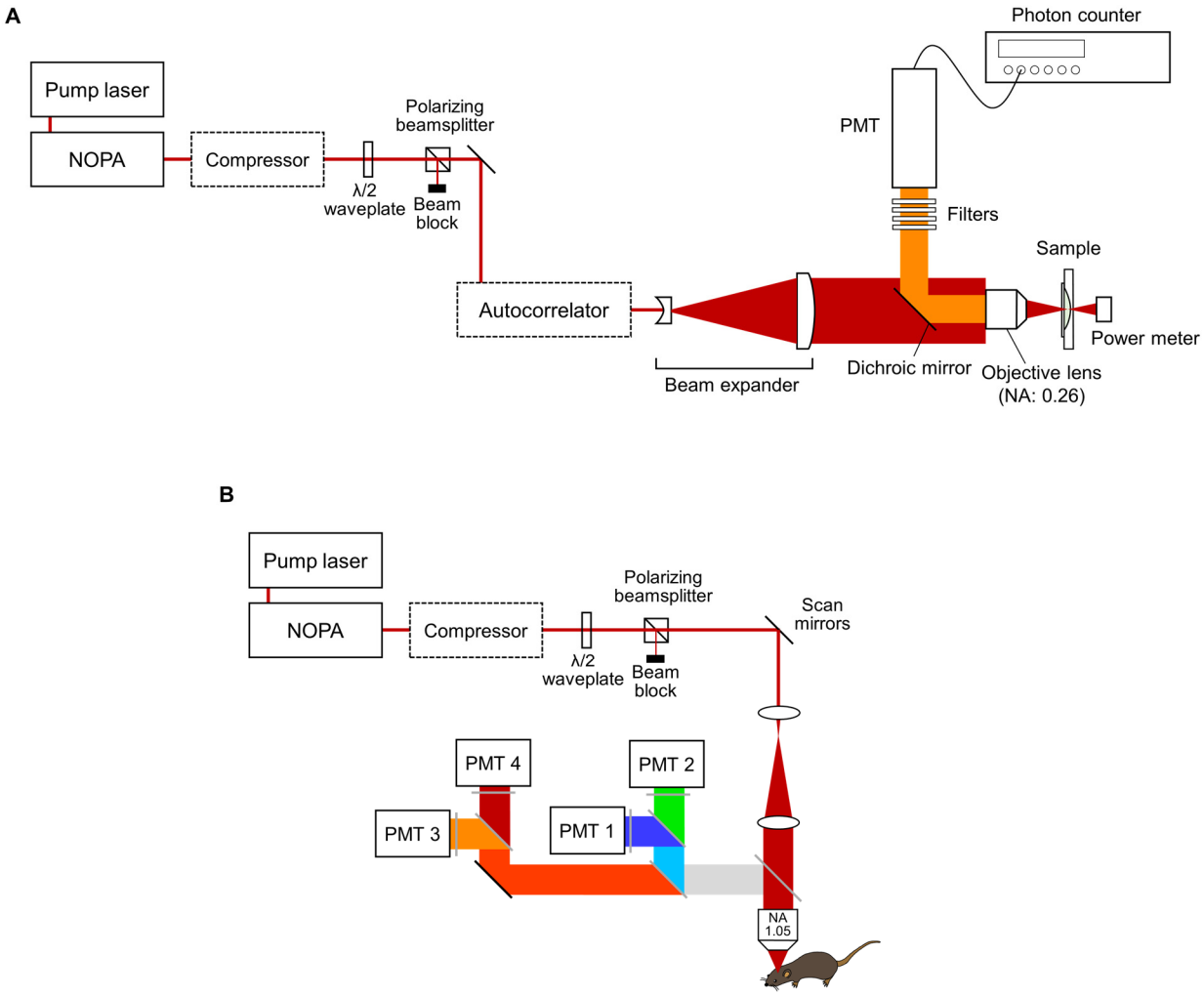
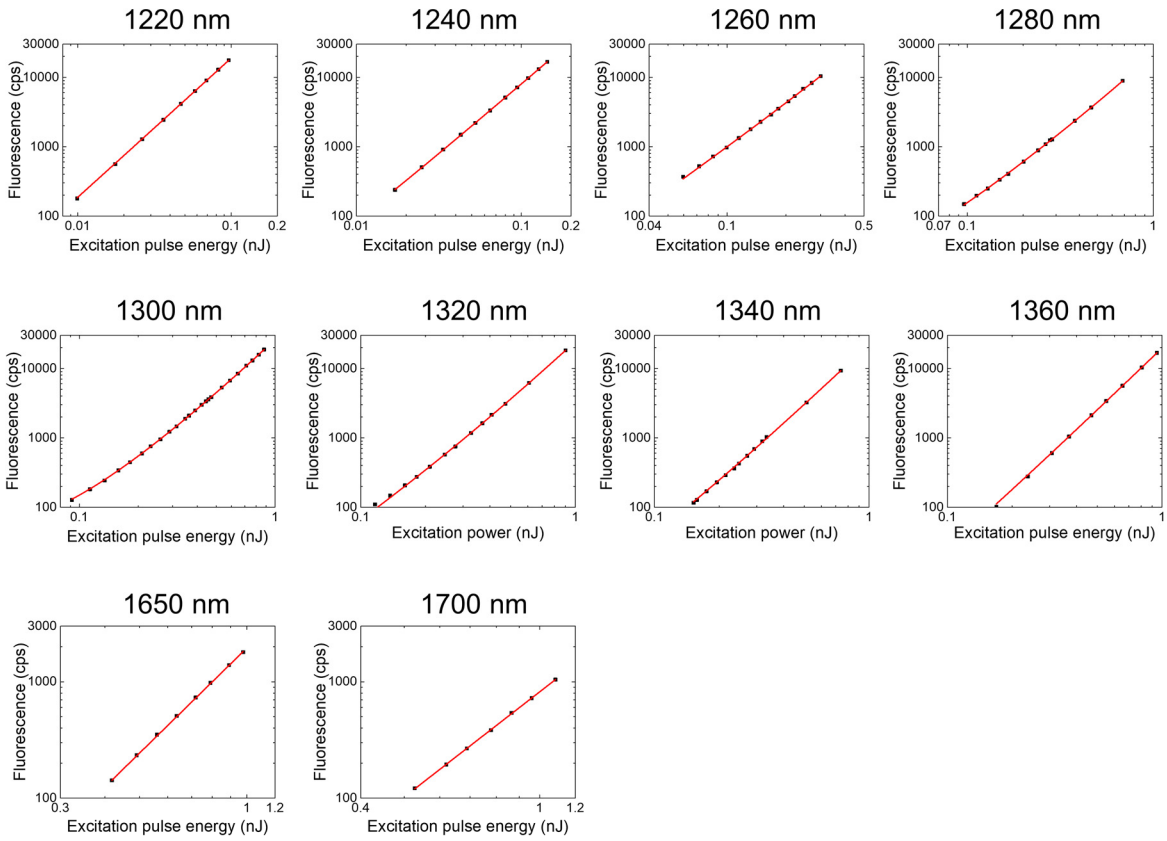


Figure S1. Experimental setups. (**A**) Multiphoton excitation cross section measurement setup. (**B**) Three-photon microscopy setup.



	1220 nm	1240 nm	1260 nm	1280 nm	1300 nm	1320 nm	1340 nm	1360 nm	1650 nm	1700 nm
A_2	1.86×10^6	7.83×10^5	9.20×10^4	1.31×10^4	9.36×10^3	4.61×10^3	2.10×10^3	6.97×10^2	---	---
A_3	---	---	7.83×10^4	8.45×10^3	1.67×10^4	1.98×10^4	2.02×10^4	1.90×10^4	1.95×10^3	8.19×10^2

Figure S2. Logarithmic plot of fluorescence signal versus excitation pulse energy at selected wavelengths for Texas Red. The black squares and the red lines show the raw data and the polynomial fits, respectively. The excitation wavelengths are indicated on the top of each plot. In the bottom table, coefficients A_2 and A_3 for the polynomial fits ($y = A_2x^2 + A_3x^3$) are shown.

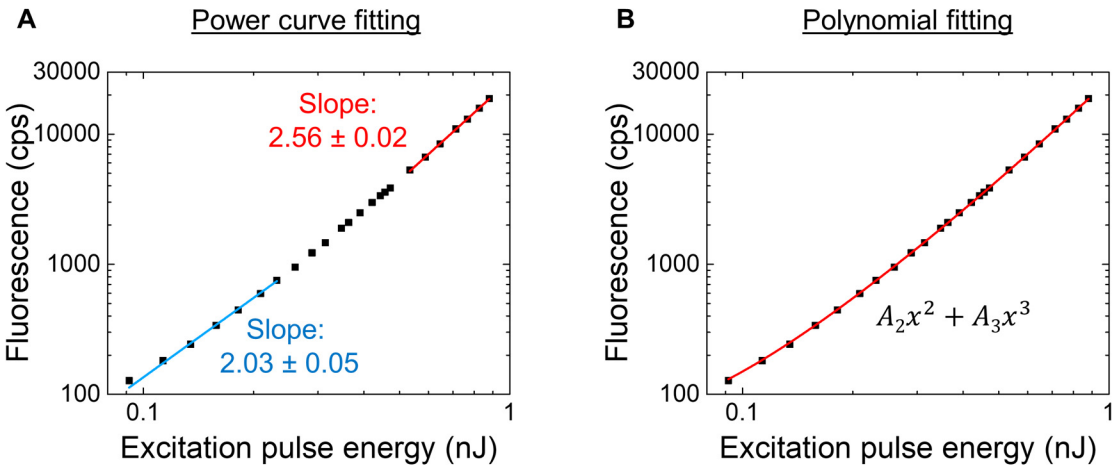


Figure S3. Logarithmic plot of Texas Red fluorescence signal versus excitation pulse energy fitted with (A) power functions at low and high power regions and (B) a polynomial function with 1300-nm excitation. Black squares and solid lines show the raw data and the fitted curves, respectively.

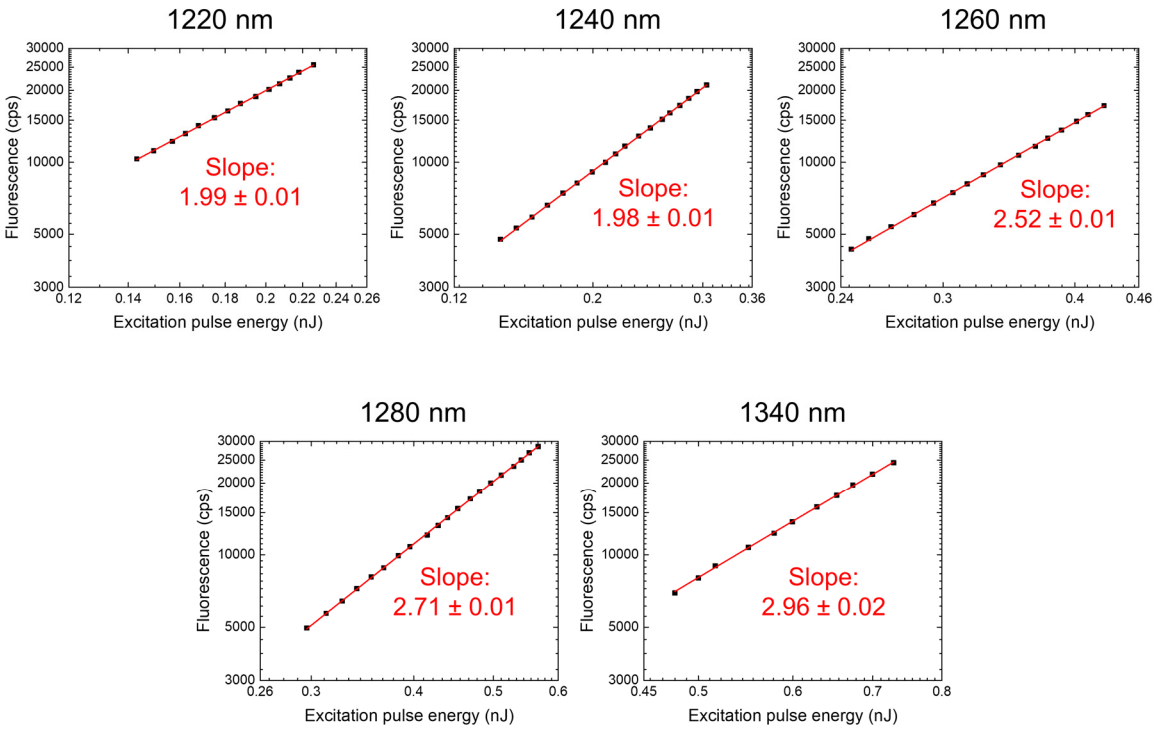


Figure S4. Texas Red fluorescence signal versus excitation pulse energy obtained with high NA (~0.8) excitation and fitted with power functions. 1 μM Texas Red in PBS solution at pH 7.4 was used. Black squares and red solid lines show the raw data and the fitted curves with power function, respectively.

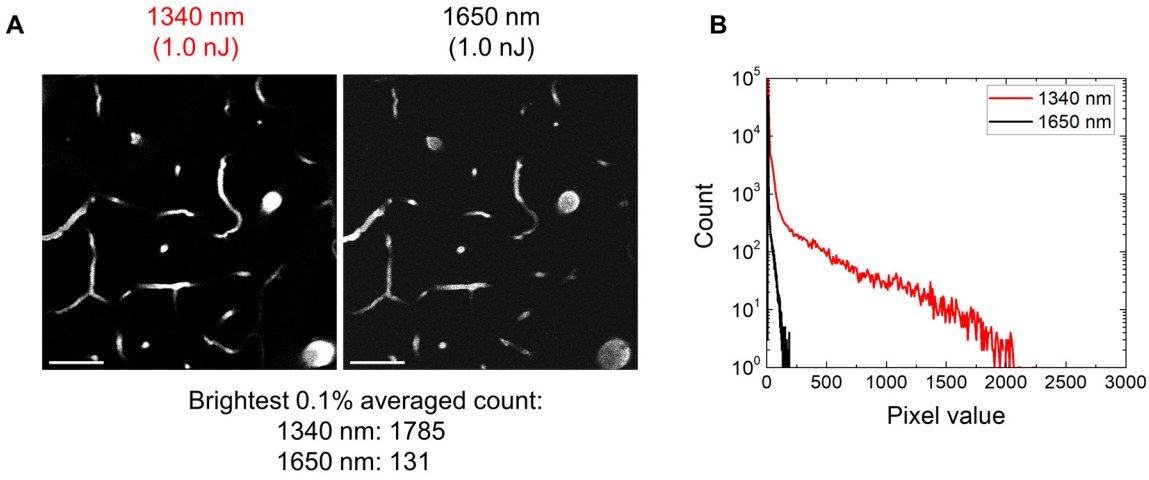
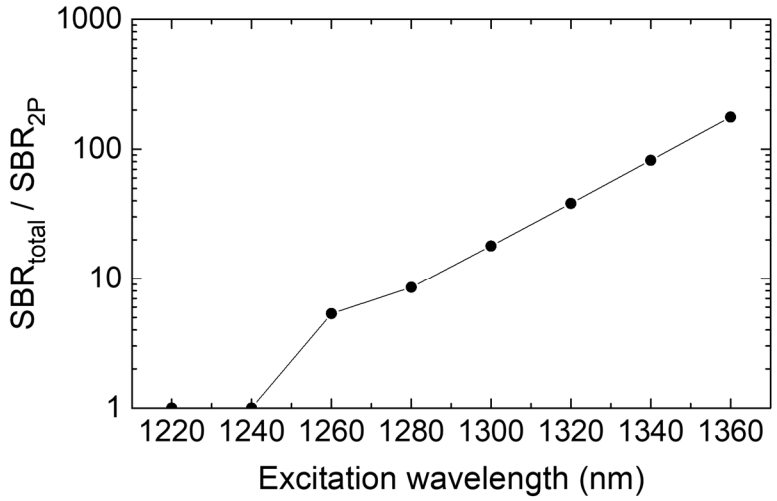


Figure S5. Multiphoton images of Texas Red labeled blood vessels in the mouse brain at 50 µm beneath the dura. 70,000 MW Texas Red dextran was injected into a wild-type mouse (C57BL/6J, female, 13 weeks old, 22 g, under anesthesia with 1.0–1.5% isoflurane in oxygen) to label the blood vessels. The images were obtained with a field-of-view of 270 × 270 µm (512 × 512 pixels per frame). The laser repetition rate was set at 330 kHz. (**A**) The images obtained with 1340 nm and 1650 nm excitation, averaged over 10 frames and 50 frames, respectively (1.0 Hz frame rate; pixel dwell time: 3.1 µs). The pulse energy at the focus of the objective lens was 1.0 nJ. The brightest 0.1% pixel counts were averaged: 1785 for 1340 nm, and 131 for 1650 nm. Scale bars, 50 µm. (**B**) The pixel intensity histograms of the images in (**A**).

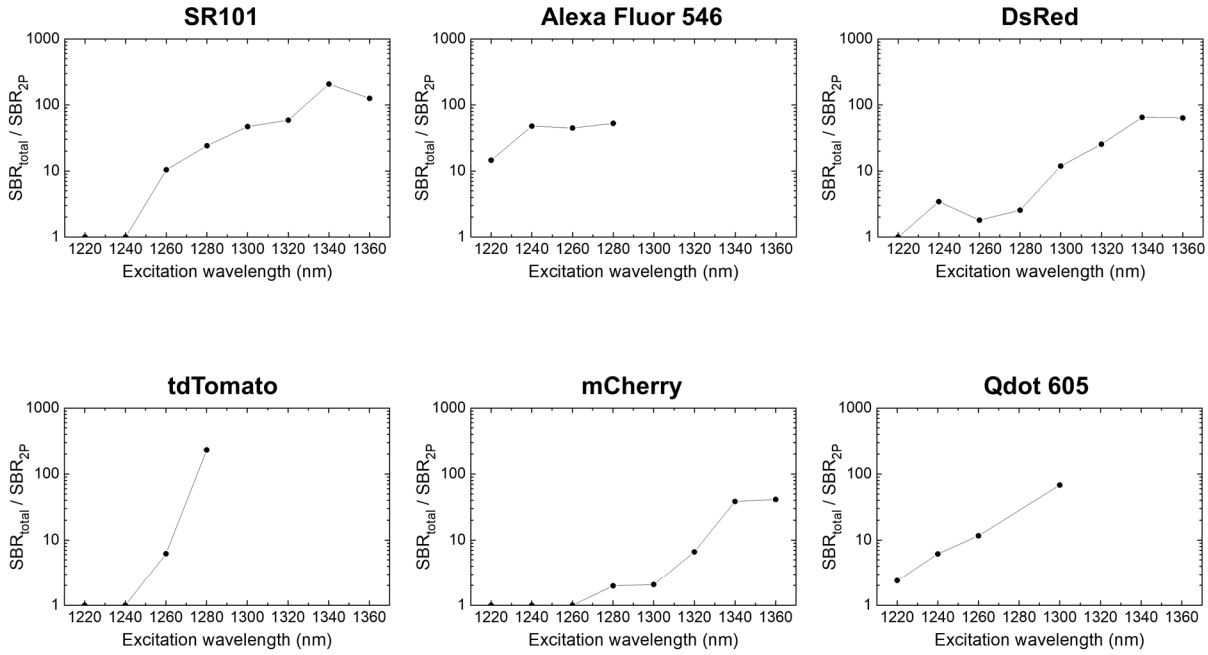
A



B

Excitation wavelength	1220 nm	1260 nm	1300 nm	1340 nm
SBR at 1150 μm exp.	0.77 (1)	1.4 (1.8)	12 (15)	47 (61)
SBR_{total}/SBR_{2P} calc.	1	5.4	18	82

Figure S6. Calculated signal-to-background ratios (SBRs) of Texas Red at selected excitation wavelengths. (**A**) Calculated SBR_{total}/SBR_{2P} for pulse energies at the focus of 0.6 nJ at 1260 nm and 1.0 nJ at all other wavelengths, with a pulse duration of 60 fs and NA of 0.8. The line is used to connect the calculated data points. See Supplemental Note 1 for the calculation. (**B**) Experimentally measured SBRs at 1150 μm depth in mouse brain (from **Fig. 2E**) and calculated SBR_{total}/SBR_{2P} from (**A**). The values inside the parenthesis are normalized to the values of 2PE at 1220 nm.



Excitation pulse energy used for the SBR calculation (nJ)	1220 nm	1240 nm	1260 nm	1280 nm	1300 nm	1320 nm	1340 nm	1360 nm
SR 101	0.2	0.4	0.6	1	1	1	1	1
Alexa Fluor 546	1	1	1	1	---	---	---	---
DsRed	1	1	1	1	1	1	1	1
TdTomato	1	1	1	1	---	---	---	---
mCherry	1	1	1	1	1	1	1	1
QD 605	0.05	0.05	0.05	---	0.05	---	---	---

Figure S7. Calculated SBR_{total}/SBR_{2P} of SR 101, Alexa Fluor 546, DsRed, tdTomato, mCherry and Qdot 605 (see Supplemental Note 1). The pulse duration is 60 fs, and the NA is 0.8. The lines are used to connect the calculated data points. The pulse energies at the focus for the calculations are listed in the table.

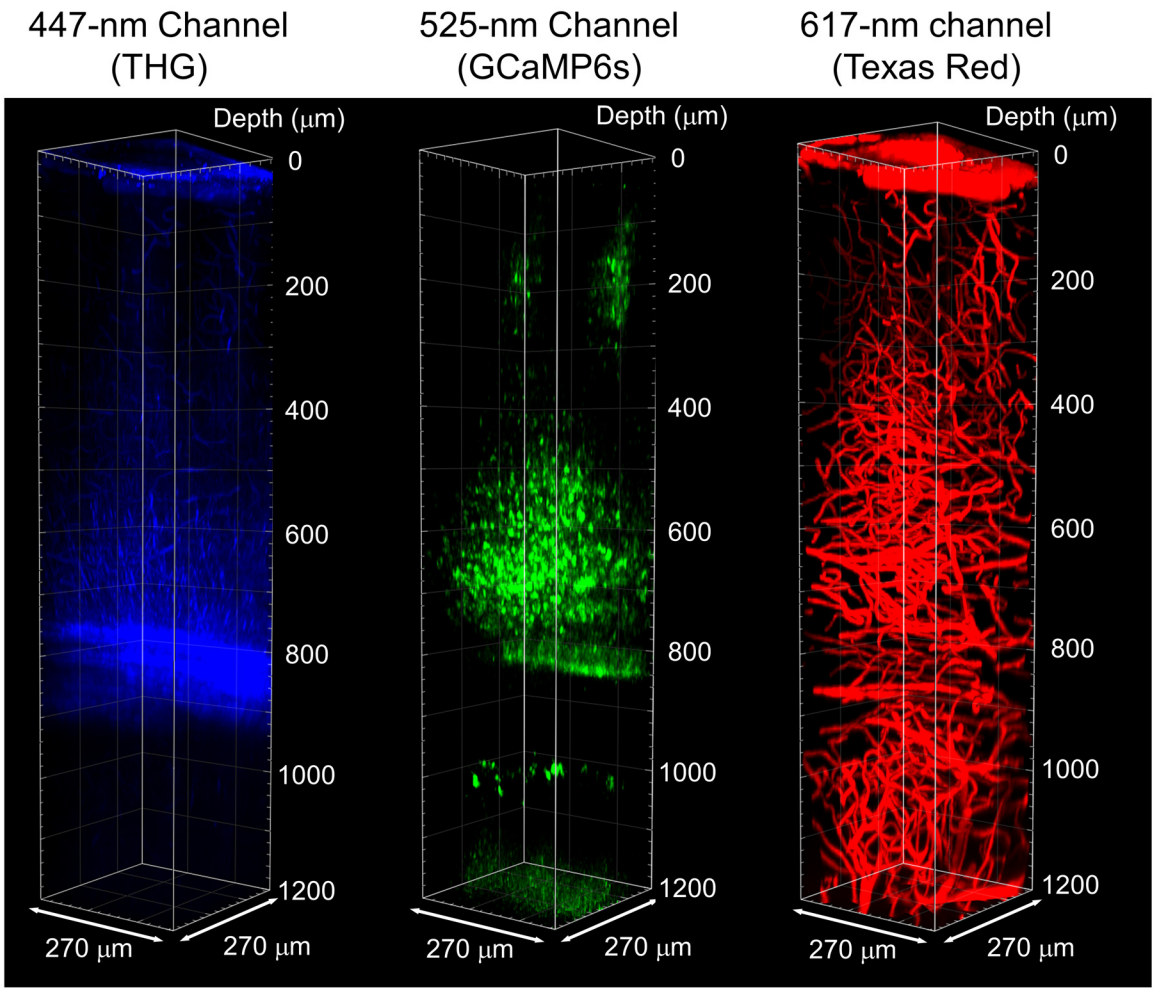


Figure S8. Separate 3D stacks for data shown in Fig. 3C. The experiments were performed simultaneously with three-channel detection.

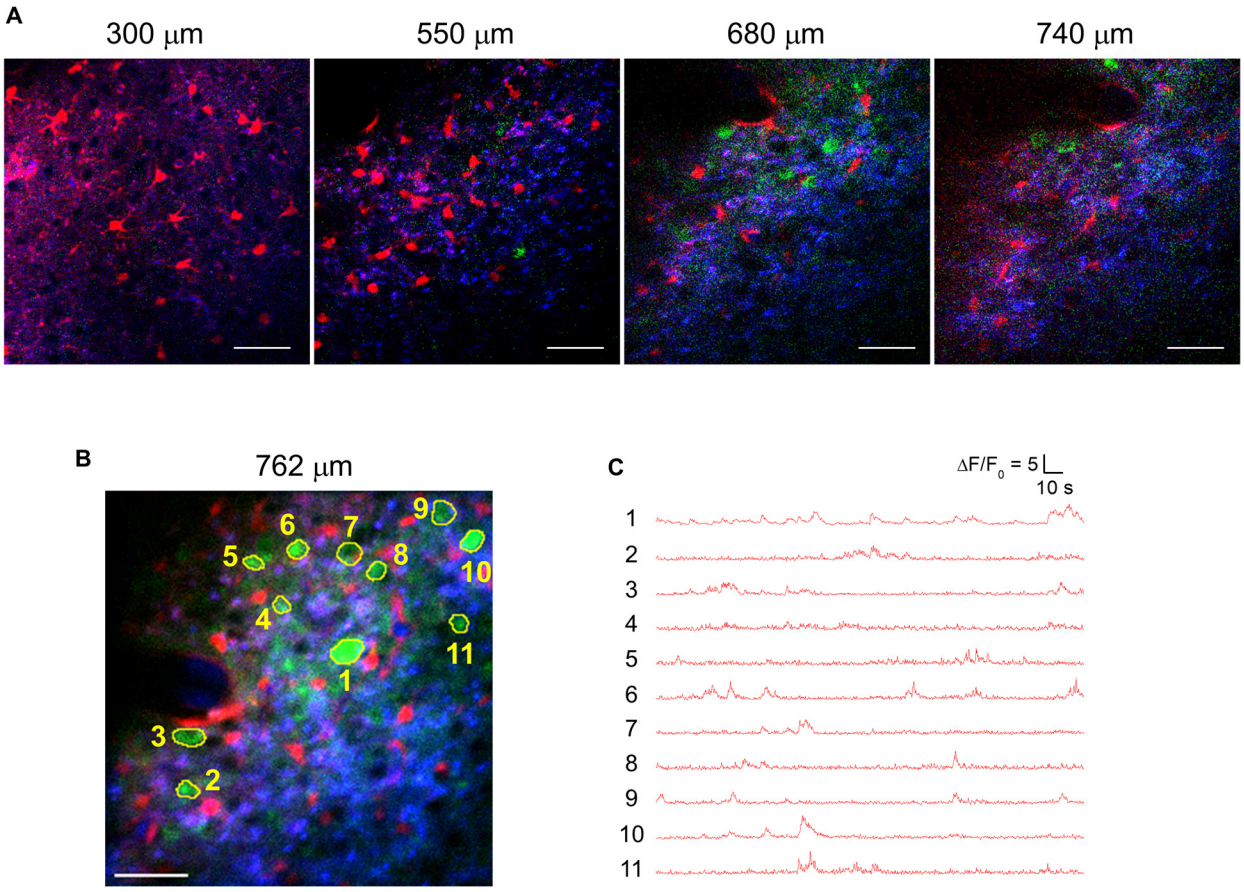


Figure S9. Multi-color three-photon fluorescence imaging of GCaMP6s-labeled neurons (green), SR 101-labeled astrocytes (red) and third harmonic generation (blue) upon 1340 nm excitation. A GCaMP6s transgenic mouse (CamKII-tTA/tetO-GCaMP6s, female, 8 weeks old, 18 g) with SR 101-labeled astrocytes was used. (**A**) Selected XY images at various depths under anesthesia with 1.0 to 1.5 % isoflurane in oxygen. The laser repetition rate was set at 2 MHz, and the images were obtained with a field-of-view of 270 × 270 μm (512 × 512 pixels per frame). The images were acquired with a frame rate of 3.3 Hz with a pixel dwell time of 0.51 μs and averaged over 20 frames at each depth. The pulse energy was ~1.5 nJ at the focus. (**B**) Activity recording site at 762 μm beneath the dura with a field-of-view of 270 × 270 μm (256 × 256 pixels). (**C**) Spontaneous activity traces from the labeled neurons indicated in (**B**). 0.5 to 1.0 % isoflurane in oxygen was applied during the activity recording. The frame rate was 8.3 Hz, and the pixel dwell time was 0.51 μs. The laser repetition rate was set at 2 MHz, and the average power under the objective lens was 70 mW. Each trace was normalized to its baseline and low-pass filtered using a hamming window of 0.72-s time constant. The entire recording session is shown in Movie S3. Scale bars, 50 μm.

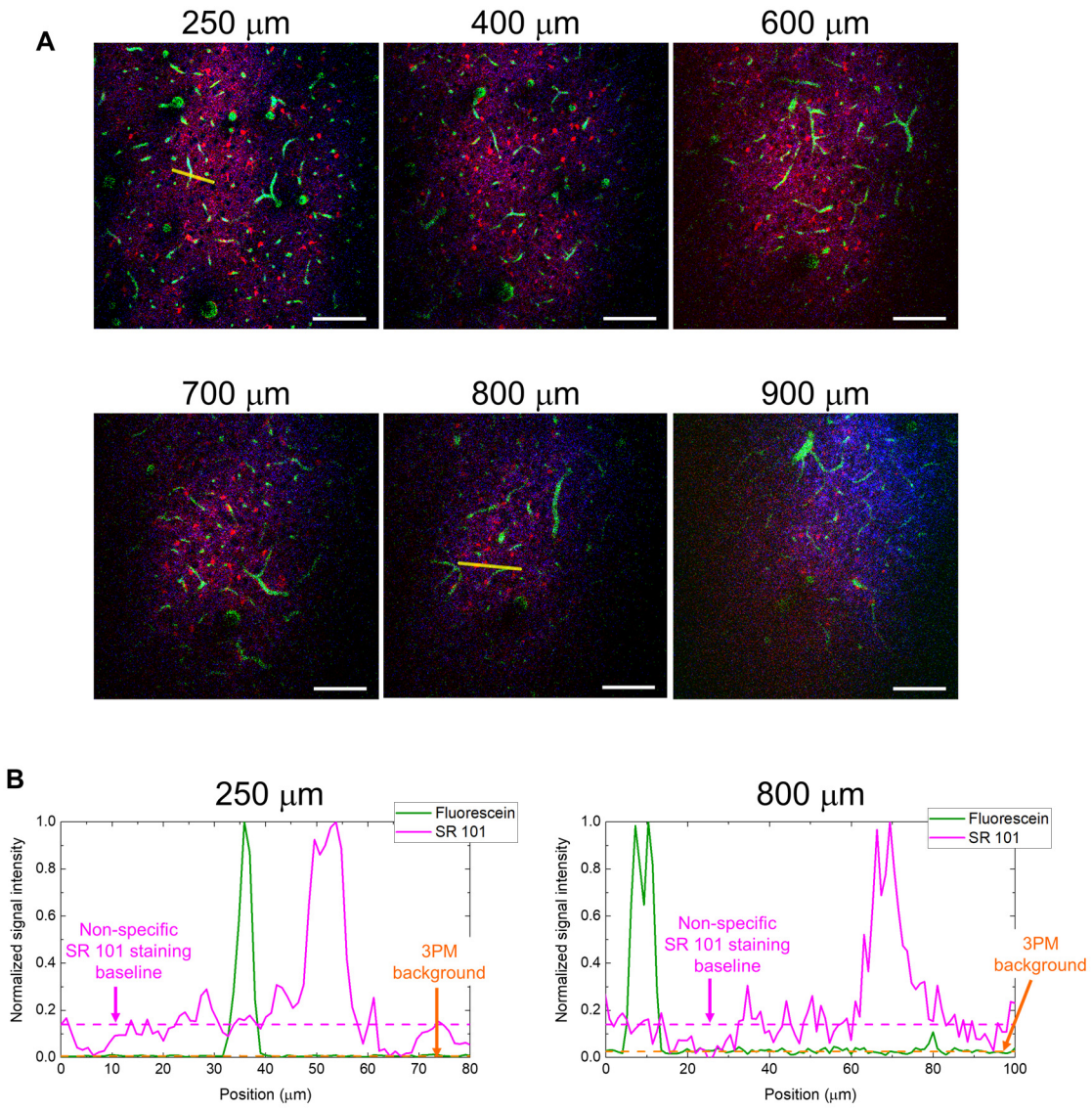


Figure S10. Multi-color three-photon imaging of fluorescein-labeled blood vessels (green), SR 101-labeled astrocytes (red) and third-harmonic generation (blue) upon 1340 nm excitation. 70,000 MW fluorescein dextran was injected into a wild-type mouse (C57BL/6J, male, 9 weeks old, 27 g, under anesthesia with 1.0 to 1.5 % isoflurane in oxygen). The images were obtained with a field-of-view of 540 × 540 μm (512 × 512 pixels per frame). (**A**) Selected 3P images at different depths. At 250–800 μm depth, the laser repetition rate was set at 2 MHz and the images were acquired with a frame rate of 3.3 Hz with a pixel dwell time of 0.51 μs (averaged over 4 frames at 250 and 400 μm, 8 frames at 600 and 700 μm, and 12 frames at 800 μm). At 900 μm depth, the laser repetition rate was set at 330 kHz and the images were acquired with a frame rate of 1.0 Hz with a pixel dwell time of 1.0 μs (averaged over 40 frames). The pulse energy was ~1.5 nJ at the focus. Scale bars, 100 μm. (**B**) The fluorescence intensity profiles at 250 μm and 800 μm depths along the yellow lines shown in **A**. The green and magenta lines show line profiles of fluorescein and SR 101, respectively. The dashed orange and magenta lines show the 3PM out-of-focus background and non-specific staining baseline of SR 101, respectively.

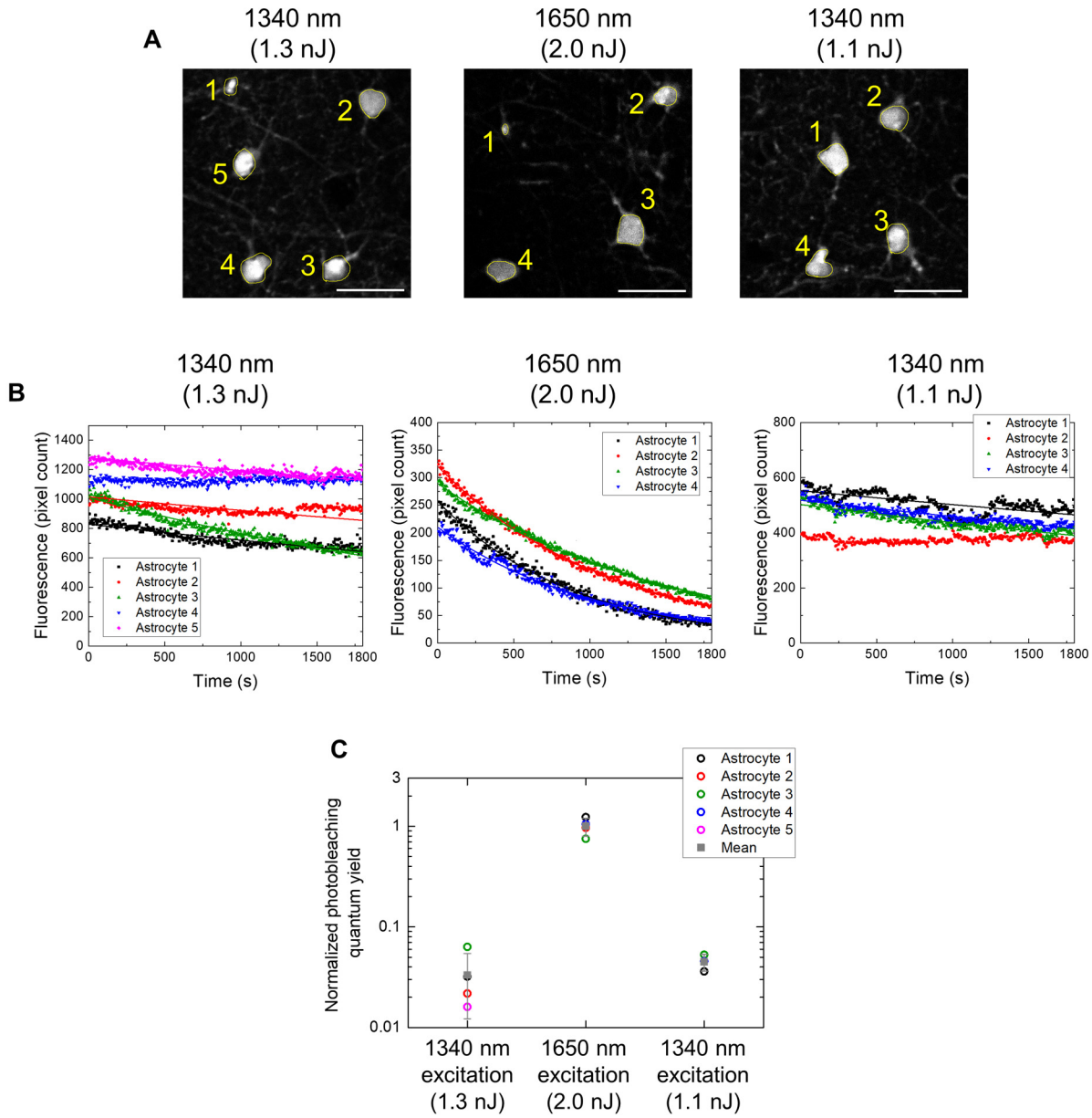


Figure S11. Photobleaching measurements of SR 101 in mouse brain with 1340- and 1650-nm excitation. A wild-type mouse (C57BL/6J, female, 35 weeks old, 24 g, under anesthesia with 1.0–1.5% isoflurane in oxygen) with SR101-labeled astrocytes was used. The images were obtained at 60 µm beneath the dura with a field-of-view of 67 × 67 µm (512 × 512 pixels per frame), a frame rate of 0.20 Hz, and a pixel dwell time of 17.9 µs. The laser repetition rate was 2 MHz. The images were obtained continuously up to 1800 s in each experiment. Unlike the results in the Fig. 5, the FOVs at 1340 nm and 1650 nm were not the same, *i.e.*, different astrocytes were imaged at 1340 nm and 1650 nm in the same mouse. A measurement with 1340-nm excitation (with ~1.3 nJ at the focus) was performed first, and 1650-nm excitation (with ~ 2.0 nJ at the focus) followed. After the measurement with 1650-nm excitation, the measurement with 1340-nm excitation (with ~1.1 nJ at the focus) was repeated. (**A**) Photobleaching measurement site at ~60 µm beneath the dura. Scale bar: 20 µm. (**B**) Photobleaching time traces from the labeled astrocytes indicated in **A**. The solid markers and lines show, respectively, the raw data

and fitted curves with single-exponential functions. For astrocyte 4 in the 1340-nm measurement with 1.3 nJ and astrocyte 2 in the 1340-nm measurement with 1.1 nJ, curve fitting was not performed because the signal did not show measurable decrease within the experiment time of 1800 seconds. (**C**) Normalized photobleaching quantum yields for each astrocyte. All data were normalized to the average value of the measurements at 1650 nm excitation. The y-axis is shown in a logarithmic scale. The open dots indicate the normalized photobleaching quantum yields for each astrocyte. The solid squares and the error bars show the mean values and the standard deviations, respectively.

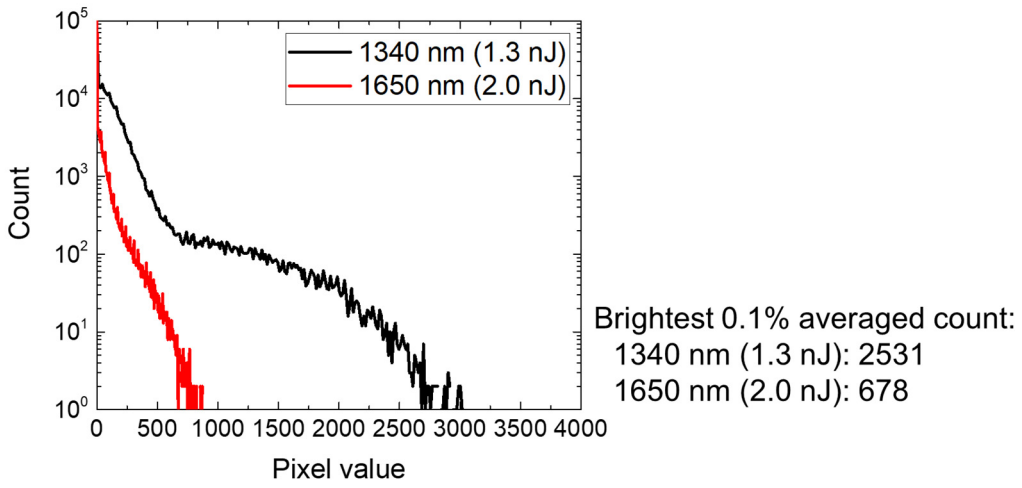
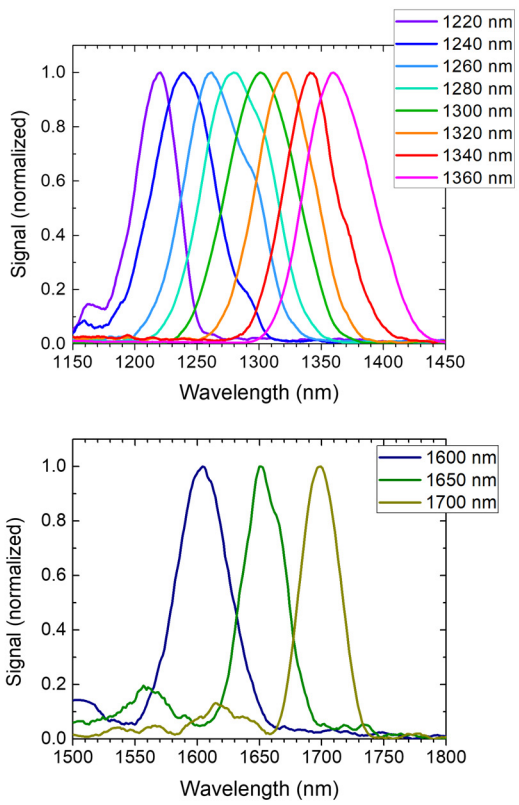


Figure S12. Pixel intensity histograms of SR 101 labeled astrocytes in mouse brain with 1340- and 1650-nm excitation at 60 µm beneath the dura. Histograms of images in Fig. S11A before photobleaching with 1340 nm excitation and 1650 nm excitation. The brightest 0.1% pixel count average with 1340 nm excitation (1.3 nJ after the objective) was 3.7 times larger than that with 1650 nm excitation (2.0 nJ after the objective). Using the 3PE cross sections in Fig. 1, the excitation power under the objective, the pulse durations, and the expected attenuation lengths (300 and 400 µm for 1340 and 1650 nm, respectively), the fluorescence signal with 1340-nm excitation is expected to be ~4.8 times larger than that with 1650-nm excitation, which is in close agreement with that measured from the *in vivo* experiments.

A



B

Wavelength	Pulse duration
1220 nm	66 fs
1260 nm	66 fs
1300 nm	41 fs
1340 nm	59 fs (50 fs for Figs. 5, S11 and S12)
1650 nm	55 fs (59 fs for Figs. 5, S11 and S12)

Figure S13. Laser spectra and pulse durations. (**A**) Measured laser spectra for both cross-section measurements and three-photon imaging. (**B**) Pulse durations measured at the focus of the objective lens for three-photon imaging, assuming hyperbolic-secant-squared pulse intensity profile. Pulse durations in the parenthesis were for the experiments in Figs. 5, S11 and S12. For all other experiments, pulse widths were listed in the table without the parenthesis.

Movie S1. Z-stack images for data shown in Fig. 3C: multi-color three-photon fluorescence imaging of GCaMP6s-labeled neurons (green), Texas Red-labeled blood vessels (red) and third harmonic generation (blue) upon 1340 nm excitation. The GCaMP6s signals (green) are shown down to 1048 μm.

Movie S2. Brain activity movie for data shown in Fig. 3E,F: multi-color three-photon imaging of GCaMP6s-labeled neurons (green), Texas Red-labeled blood vessels (red) and third harmonic generation (blue) upon 1340 nm excitation.

Movie S3. Brain activity movie for data shown in Fig. S9B,C: multi-color three-photon fluorescence imaging of GCaMP6s-labeled neurons (green), SR 101-labeled astrocytes (red) and third harmonic generation (blue) upon 1340 nm excitation.